

LAPR3 2025/2026

Logistics On Rails – Sprint 3

G103

Group members:

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Sprint planning – Sprint 3

Planned Start Date: 24/11

Real Start Date: 03/12

Planned Finished Date: 04/01

Real Finish Date: 04/01

SPRINT BACKLOG

Number of stories	24	
Number of bugs	0	
Number of Sub-Tasks:	51	
Number of Management Tasks (Scrum)	4	
Number of team members	Planned: 5	Actual: 5
Total planned estimation vs execution	Planned hours: 234 Executed hours: 219	

Sprint planning

Assignment US (as stated on project assignment)	Committed? (Yes / no according to team planning)	Todo	Doing	Testing	Done	Blocked
USAC10	Yes				X	
USAC11	Yes				X	
USAC12	Yes				X	
USAC13	Yes				X	
USAC14	Yes				X	
USAC15	Yes				X	
USAC16	Yes				X	
USBD31	Yes				X	
USBD32	Yes				X	
USBD33	Yes				X	

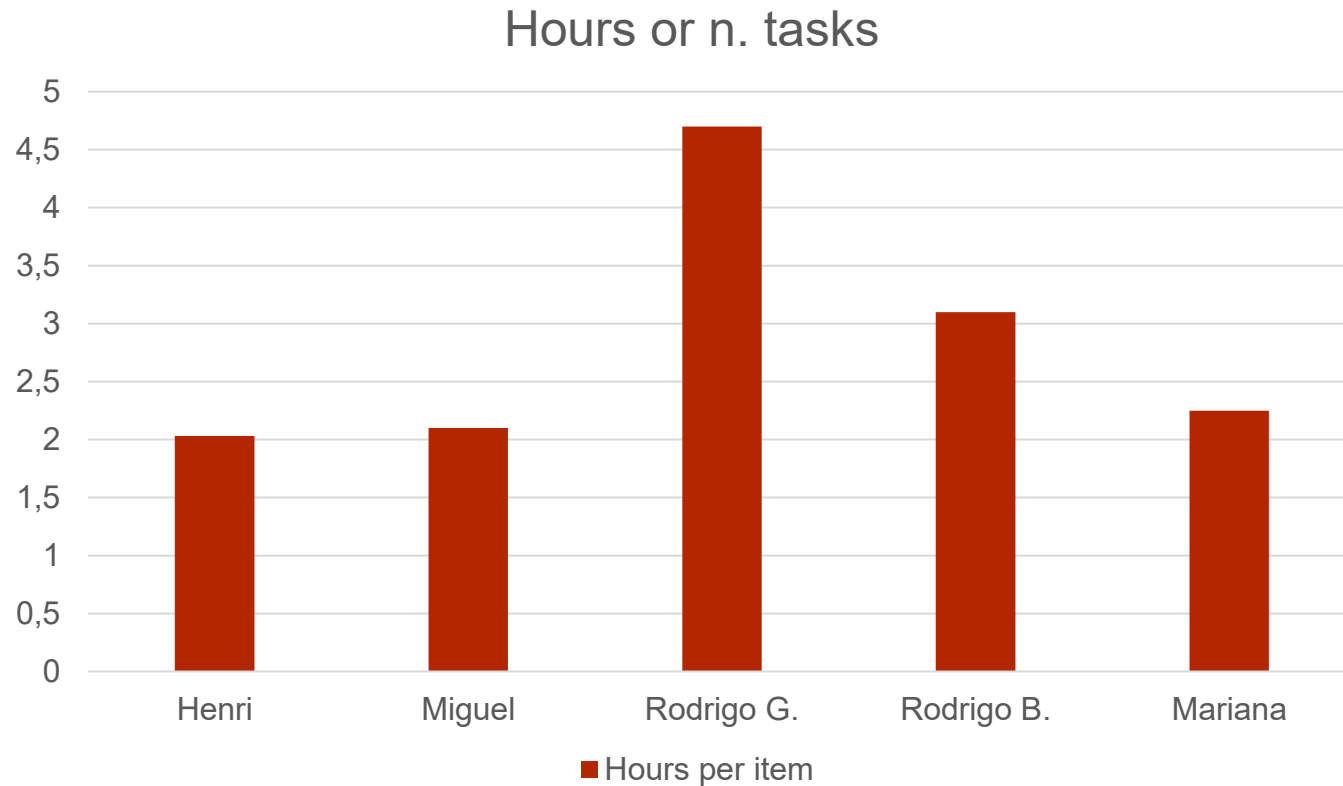
Sprint planning

Assignment US (as stated on project assignment)	Committed? (Yes / no according to team planning)	Todo	Doing	Testing	Done	Blocked
USBD35	Yes				X	
USBD36	Yes				X	
USBD38	Yes				X	
USBD40	Yes				X	
USBD43	Yes				X	
USBD44	Yes				X	
USLP08	Yes				X	
USLP09	Yes				X	
USLP10	Yes				X	
USEI11	Yes				X	

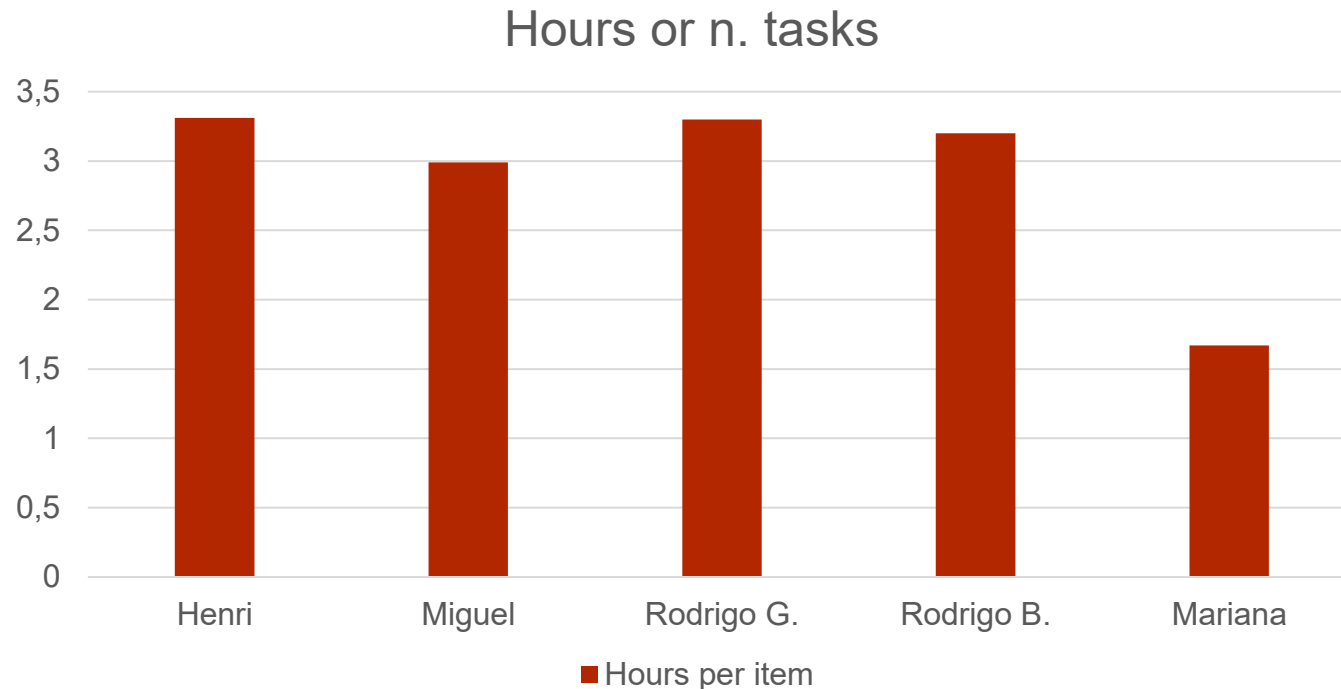
Sprint planning

Assignment US (as stated on project assignment)	Committed? (Yes / no according to team planning)	Todo	Doing	Testing	Done	Blocked
USEI12	Yes				X	
USEI13	Yes				X	
USEI14	Yes				X	
USEI15	Yes				X	

Work by team member (Sprint 3)



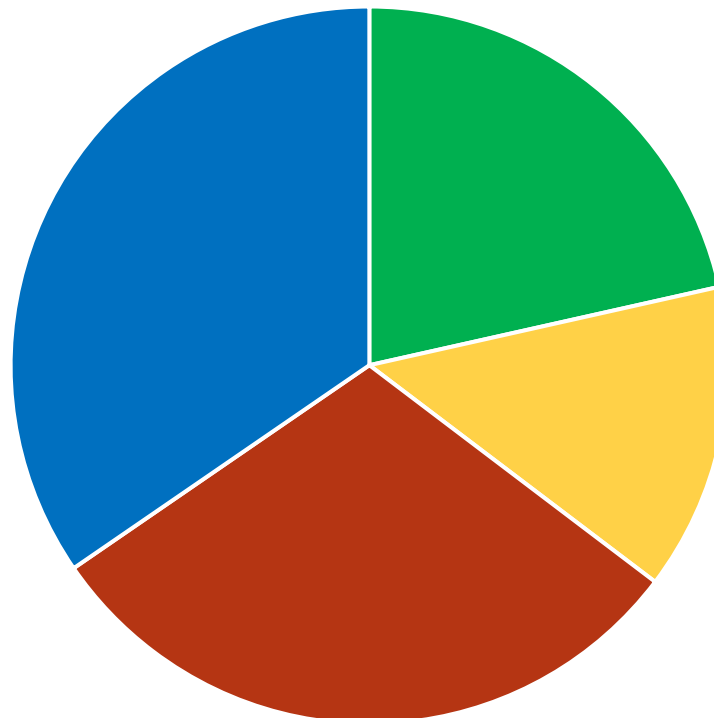
Work by team member (All Sprints)



Brief justification: The equality in our working hours is visible through the graphic. The differences appears when sometimes we created more tasks and splitted the work more, sometimes not. In general, the work was balanced between the five teammembers.

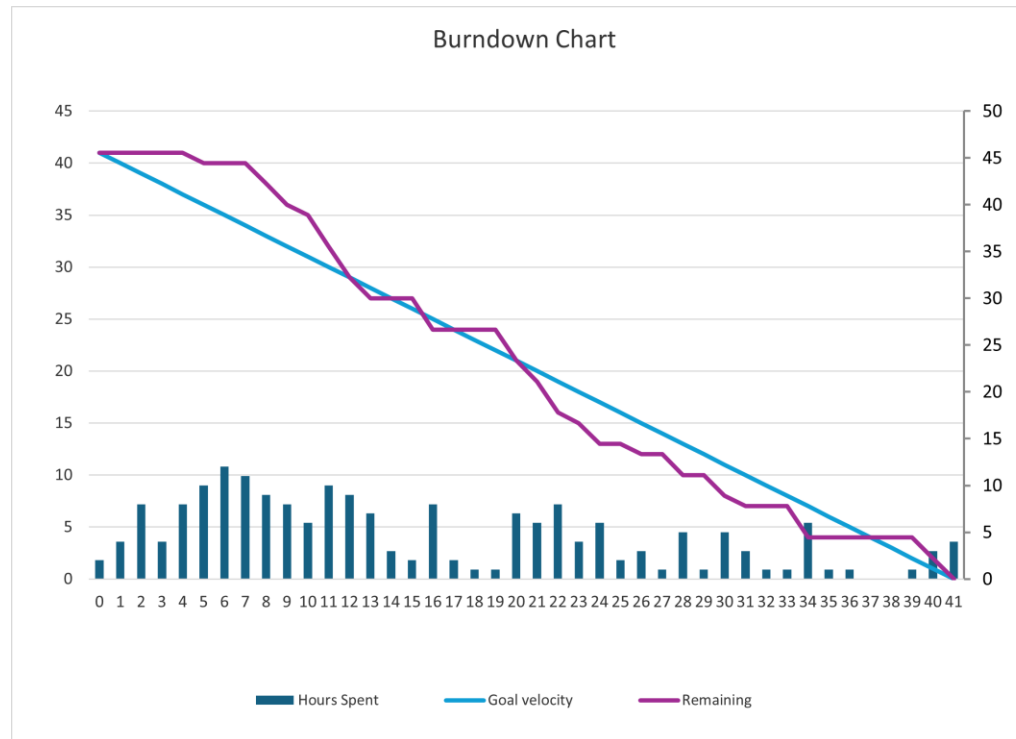
Work by type

Epic or Component or UC



■ ESINF ■ ARQCP ■ LAPR3 ■ BDDAD

Sprint Burndown Chart



Brief comment: Our team worked more intensively during the middle-ending of the sprint. Although not having a consistent line of work during the whole sprint, we worked hard, and effectively and also we were able to finish everything on time.

Issues / Assumptions / unanswered questions

Description	Responsible	Date / sprint	Related issue / item in Git
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-

Daily Meetings (Sprint 3)

Date	Place / duration	Mariana	Rodrigo B.	Rodrigo G.	Henri	Miguel
3/12/2025	LAPR3 Class / 60 min	x	x	x	x	
4/12/2025	LAPR3 Class / 30 min	x	x	x	x	x
5/12/2025	ISEP / 20min	x	x	x	x	x
9/12/2025	ISEP / 30min	x	x	x	x	x
10/12/2025	LAPR3 Class / 30 min	x	x	x	x	x
11/12/2025	LAPR3 Class / 30 min	x	x	x	x	x
12/12/2025	LAPR3 Class / 30 min	x	x	x	x	x
15/12/2025	ISEP / 20 min	x	x	x	x	x

Daily Meetings (Sprint 3)

Date	Place / duration	Mariana	Rodrigo B.	Rodrigo G.	Henri	Miguel
16/12/2025	ISEP / 20 min	x	x	x	x	x
17/12/2025	LAPR3 Class / 20 min	x	x	x	x	x
18/12/2025	ISEP / 20min	x	x	x	x	x
19/12/2025	ISEP / 20 min	x	x	x		x
24/12/2025	Discord / 60 min	x	x	x	x	x
27/12/2025	Discord / 60 min	x	x	x	x	x
29/12/2025	Discord / 60 min	x	x	x	x	x
2/1/2026	Discord / 60 min	x	x	x	x	x
4/1/2026	Discord / 60 min	x	x	x	x	x

LLM assessment (ChatGPT, Copilot, or other)

	With daily meeting records	With daily meeting records, status report and project assignment		
LLM Evaluation (0-10):	8/10	9/10	Team Evaluation (0-10):	7.5/10
LLM team and project assessment summary output:	Team score: 9/10. Meetings reflect iterative progress matching the MVP sprints (e.g., domain/relational models in Sprint 1, assembly functions in Sprint 2, optimizations like max flow and centrality in Sprint 3), with collaboration on dependencies (e.g., database integration, sensors) and tools like GitHub Boards/Visual Paradigm. Completion of TDD, PL/SQL, graphs, and hardware (e.g., Raspberry Pi Pico) shows solid adherence to requirements like agile SCRUM, TDD, and repository structure.			

Comments on LLM Evaluation:	We agree with the LLM opinion over our work, but we know that taking into account only our daily meetings, status report and the project assignment is not enough to really analyze our work during the sprints. Despite that, we think it's matching our reality, since we've been improving a lot since the beginning of the project.	Team valuation on LLM comments (Very Good, Good, Medium, Bad)::	Good
	LLM improvement ideas		
Ask LLM for 4 improvements ideas:	Enhance Log Consistency: Standardize formats (e.g., always list all participants, quantify progress in hours) and add metrics like burndown charts directly in meetings for better visualization. Reduce Dependency Delays: Implement mock data/stubs earlier for blocked US (e.g., USEI10 waiting on KD-trees) and cross-review designs pre-coding. Boost Documentation Depth: Include code snippets or complexity graphs in "What I did" sections; generate UML auto from code for retrospectives. Strengthen Hardware Testing: Schedule dedicated sessions for ARQCP physical integration (e.g., sensors/LEDs) with video demos to accelerate debugging.		
Which ideas would you implement? (1, 2, 3, 4?)	2 and 3.		

LLM performance assessment

Include here as attachment a file with the input and output prompt

Final Sprint Retrospective (FLAP) Date: 05/01

Future considerations:

- For future academic and professional projects, we'll start with mock data right away to avoid getting stuck waiting on other people's tasks. Setting up automated tests early and doing more pair programming will help us learn tough topics faster. Also, regular code reviews and the same documentation format for everyone will make sharing knowledge smoother.

Lessons Learned:

- Daily meetings really helped spot problems early, especially when one task depended on another. Planning poker worked well when we broke tasks into small pieces first. Using repositories made connecting our code to the database much easier. Talking through hard concepts together sped up everyone's learning on things like FIFO logic and graph algorithms

Problem Areas:

- Dependencies between tasks held some people up. The sensor debugging took way longer than expected. We all struggled at first with assembly code and graph algorithms, which slowed Sprint 2 and 3. Some meetings had missing people, and documentation wasn't always at the same level across the team.

Accomplishments:

- We finished every User Story across all three sprints - cargo unloading with FEFO, KD-tree searches, network optimization, Raspberry Pi station controls, and all the PL/SQL functions. Our GitHub board tracked everything perfectly, and we wrote solid unit tests with complexity analysis for all components. Getting the hardware and database working together on Oracle LiveSQL was a big win.

Overall Final Project Mood:

Independently on the team performance, what's the final Team feeling:



Brief justification:

We're happy with how our team worked. We had some delays waiting for other tasks and sensor problems, but we finished everything with good teamwork and daily meetings. Getting the hard parts like sensors and math algorithms working made us proud of what we achieved.

